



Netflix Report

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08\06\2026



Objectives

The goal of this report is to explore Netflix as a market.

The focus is on identifying which films and series are most prevalent among viewers, as well as which genres and regions dominate.

This analysis will greatly contribute to the key question: Whether to engage in Netflix investment and what risks could be of critical importance.



Analysis 1: How is our data structured?

In dataset we have this columns: id, title, type, description, release_year, age_certification, runtime, genres, production_countries, seasons, imdb_id, imdb_score, imdb_votes,tmdb_popularity, tmdb_score.

	id	title	type	description	release_year	age_certification	runtime	genres	production_countries	seasons	imdb_id	imdb_score	imdb_votes	tmdb_popularity	tmdb_score
0	ts300399	Five Came Back: The Reference Films	SHOW	This collection includes 12 World War II-era p...	1945	TV-MA	51	['documentation']	['US']	1.0	NaN	NaN	NaN	0.600	NaN
1	tm84618	Taxi Driver	MOVIE	A mentally unstable Vietnam War veteran works ...	1976	R	114	['drama', 'crime']	['US']	NaN	tt0075314	8.2	808582.0	40.965	8.179
2	tm154986	Deliverance	MOVIE	Intent on seeing the Cahulawassee River before...	1972	R	109	['drama', 'action', 'thriller', 'european']	['US']	NaN	tt0068473	7.7	107673.0	10.010	7.300
3	tm127384	Monty Python and the Holy Grail	MOVIE	King Arthur, accompanied by his squire, recrui...	1975	PG	91	['fantasy', 'action', 'comedy']	['GB']	NaN	tt0071853	8.2	534486.0	15.461	7.811
4	tm120801	The Dirty Dozen	MOVIE	12 American military prisoners in World War II...	1967	NaN	150	['war', 'action']	['GB', 'US']	NaN	tt0061578	7.7	72662.0	20.398	7.600
5	ts22164	Monty Python's Flying Circus	SHOW	A British sketch comedy series with the shows ...	1969	TV-14	30	['comedy', 'european']	['GB']	4.0	tt0063929	8.8	73424.0	17.617	8.306
6	tm70993	Life of Brian	MOVIE	Brian Cohen is an average young Jewish man, bu...	1979	R	94	['comedy']	['GB']	NaN	tt0079470	8.0	395024.0	17.770	7.800
7	tm14873	Dirty Harry	MOVIE	When a madman dubbed 'Scorpio' terrorizes San ...	1971	R	102	['thriller', 'action', 'crime']	['US']	NaN	tt0066999	7.7	155051.0	12.817	7.500
8	tm119281	Bonnie and Clyde	MOVIE	In the 1930s, bored waitress Bonnie Parker fal...	1967	R	110	['crime', 'drama', 'action']	['US']	NaN	tt0061418	7.7	112048.0	15.687	7.500
9	tm98978	The Blue Lagoon	MOVIE	Two small children and a ship's cook survive a...	1980	R	104	['romance', 'action', 'drama']	['US']	NaN	tt0080453	5.8	69844.0	50.324	6.156

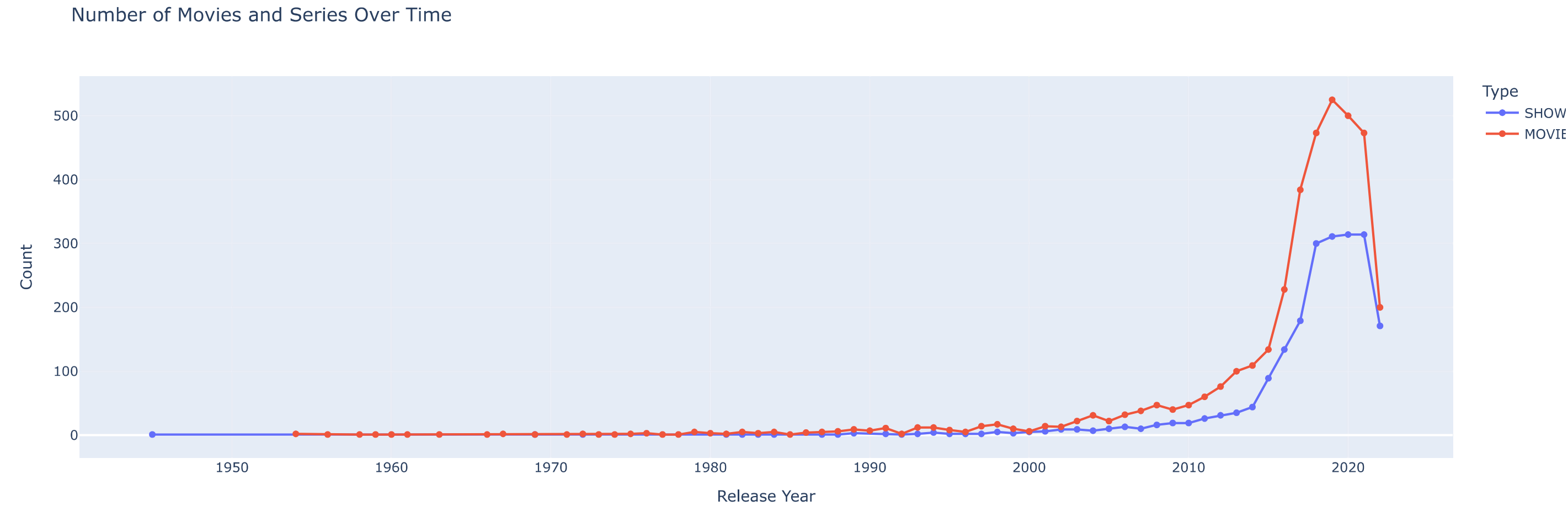
There is 5850 rows in dataset.

Analysis 2: Preparing the Dataset for Analysis: Null Value Inspection

- ✅ Column 'id' has no missing values (reliable).
- ⚠️ Column 'title' has 1 missing values (minor issue).
- ✅ Column 'type' has no missing values (reliable).
- ⚠️ Column 'description' has 18 missing values (minor issue).
- ✅ Column 'release_year' has no missing values (reliable).
- 🚨 Column 'age_certification' has 2619 missing values (major issue).
- ✅ Column 'runtime' has no missing values (reliable).
- ✅ Column 'genres' has no missing values (reliable).
- ✅ Column 'production_countries' has no missing values (reliable).
- 🚨 Column 'seasons' has 3744 missing values (major issue).
- ⚠️ Column 'imdb_id' has 403 missing values (moderate issue).
- ⚠️ Column 'imdb_score' has 482 missing values (moderate issue).
- ⚠️ Column 'imdb_votes' has 498 missing values (moderate issue).
- ⚠️ Column 'tmdb_popularity' has 91 missing values (moderate issue).
- ⚠️ Column 'tmdb_score' has 311 missing values (moderate issue).

In this dataset we have several columns with null values that are unreliable for analysis, so we will rely on those that provide accurate information, namely: type, release_year, genre, title. The columns imdb_score, tmdb_score, and tmdb_popularity contained 482 (8.2%), 311 (5.3%), and 91 (1.6%) missing values respectively. Since the proportion of missing values was relatively low and these variables were important for the analysis, the columns were retained. Missing values were excluded only from analyses and visualizations that required these specific variables.

Analysis 3: Distribution of Content and Viewing Trends using Line Chart

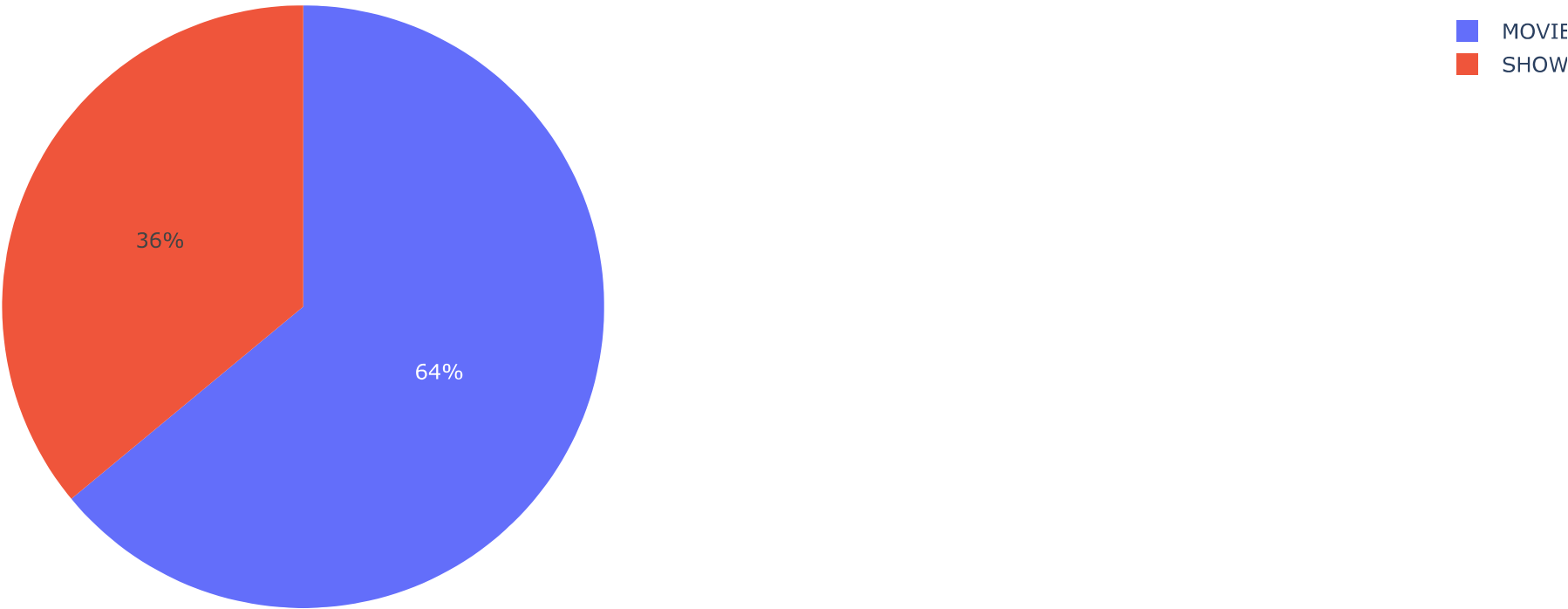


- Key Insights:
- SHOW peaked in 2020 with 314 titles
 - MOVIE peaked in 2019 with 525 titles

This chart shows how the number of movies and series has changed over the years. Older releases are fewer, while newer releases demonstrate a significant increase in both movies and series. The slight decline in 2020 indicates that the trend is not always linear, but overall it reflects a general growth.

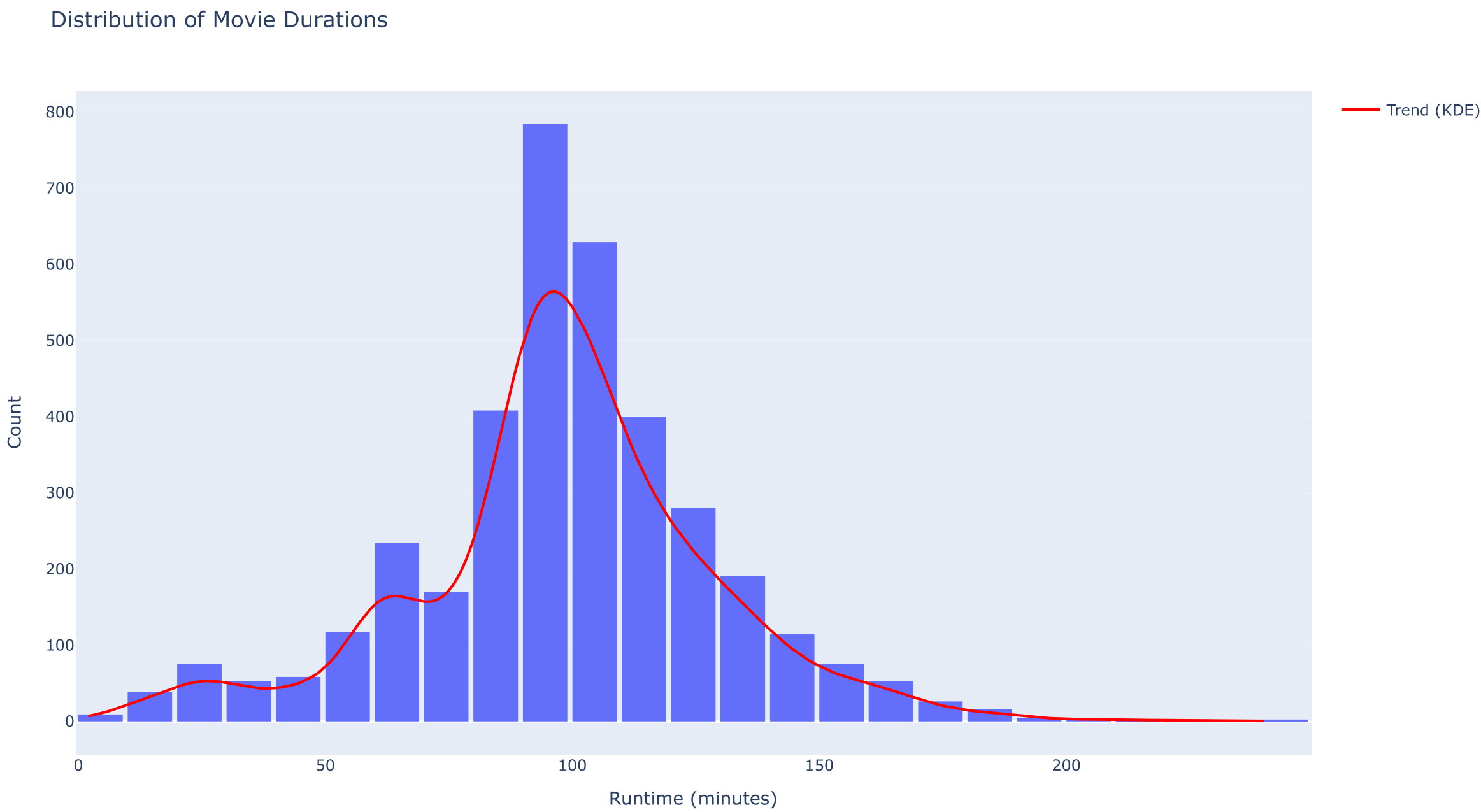
 Analysis 4: Content Structure (Movies or Shows)

Share of Movies vs Shows



The pie chart highlights the predominance of movies in the dataset, totaling 3,744 entries, compared to 2,106 series. This distribution indicates that movies represent the majority share of the content. Similar to the temporal analysis, movies continue to hold a dominant position compared to series.

 Analysis 5: Distribution of Movie Durations

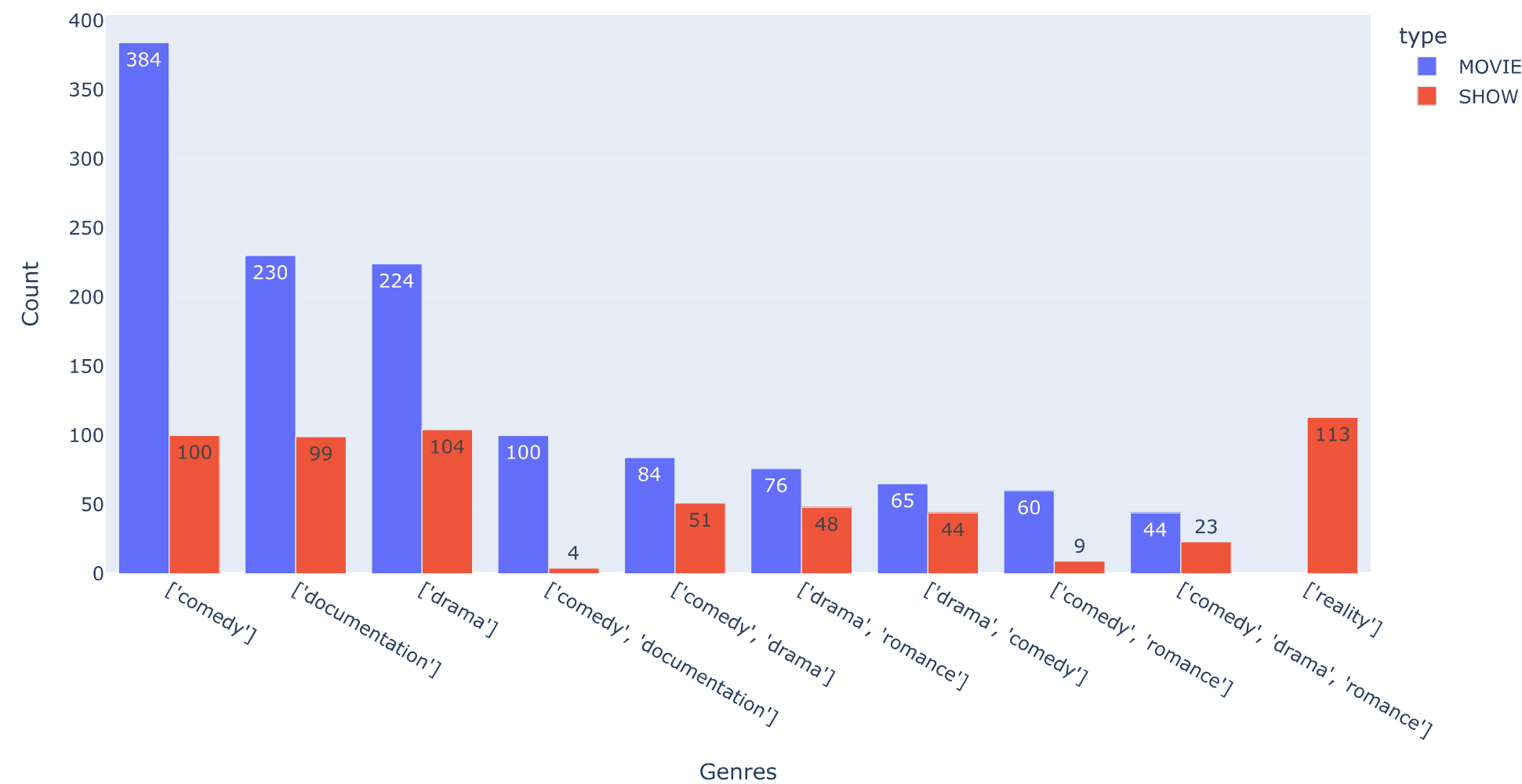


The chart indicates that most movies have a runtime of around 100 minutes, with shorter and longer films being less common. This highlights a clear central tendency in movie durations.

Analysis 6: Genre Distribution in the Dataset

Top 10 Genres Representation through Bar Chart:

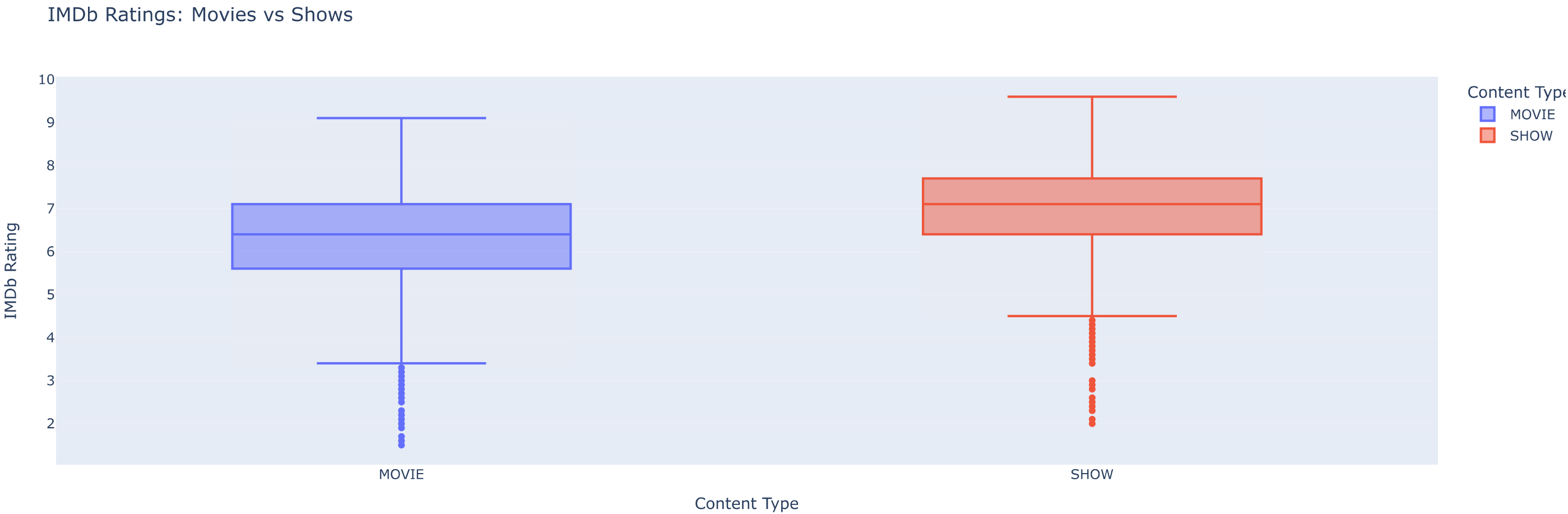
Top 10 Genres by Type   

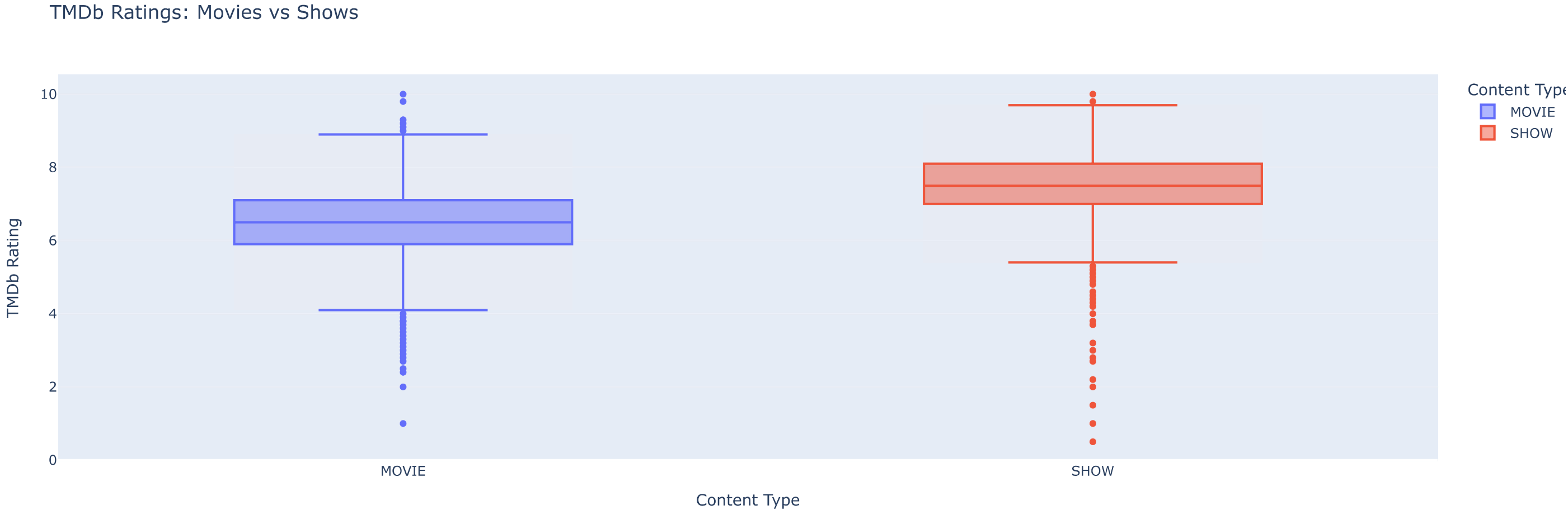


The grouped bar chart demonstrates that movies dominate genre representation, with higher counts across most categories compared to shows. The genre that stands out the most is comedy,after him are documentation and drama.

 Analysis 7: Ratings on IMDb and TMDb





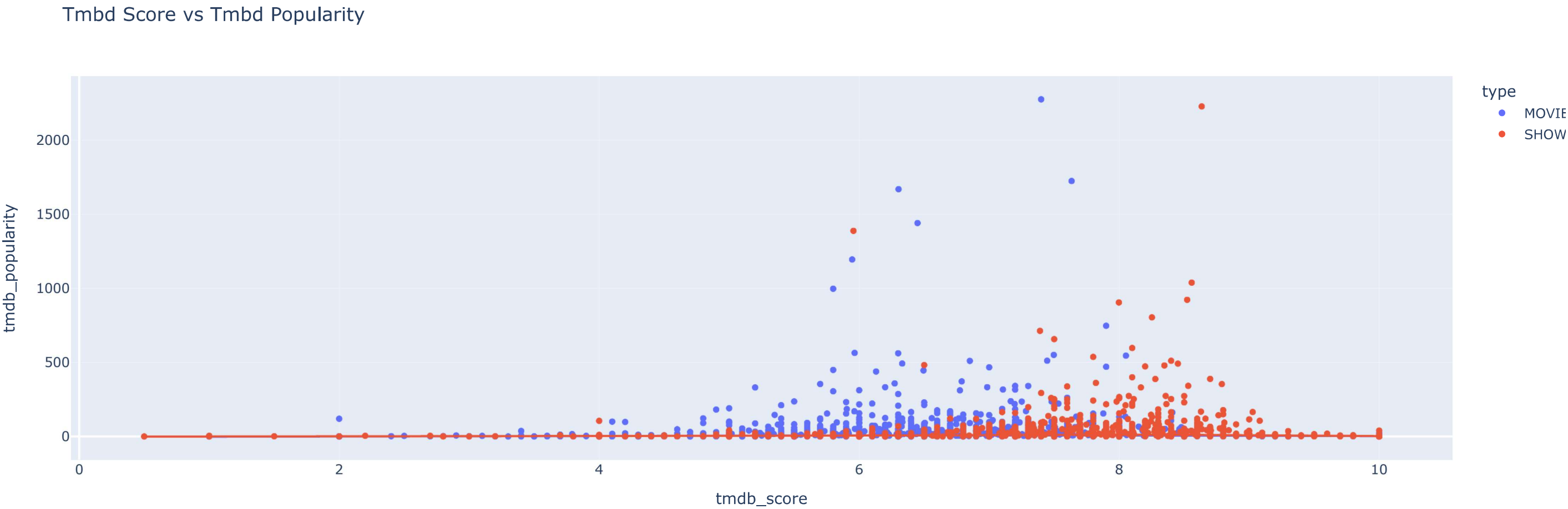
Key Insights:

- SHOW: average IMDb score = 6.98, average TMDb score = 7.48
- MOVIE: average IMDb score = 6.25, average TMDb score = 6.46

On both charts, it is clearly and surprisingly visible that shows lead in user ratings. While movies dominate in all other segments, in this case shows have taken the leading role. Although shows lead in ratings on both IMDb and TMDb, there is one movie with a perfect score of 10 that stands out from the rest.

Analysis 8: Relationship between Ratings and Popularity

Tmbd Score vs Tmbd Popularity..

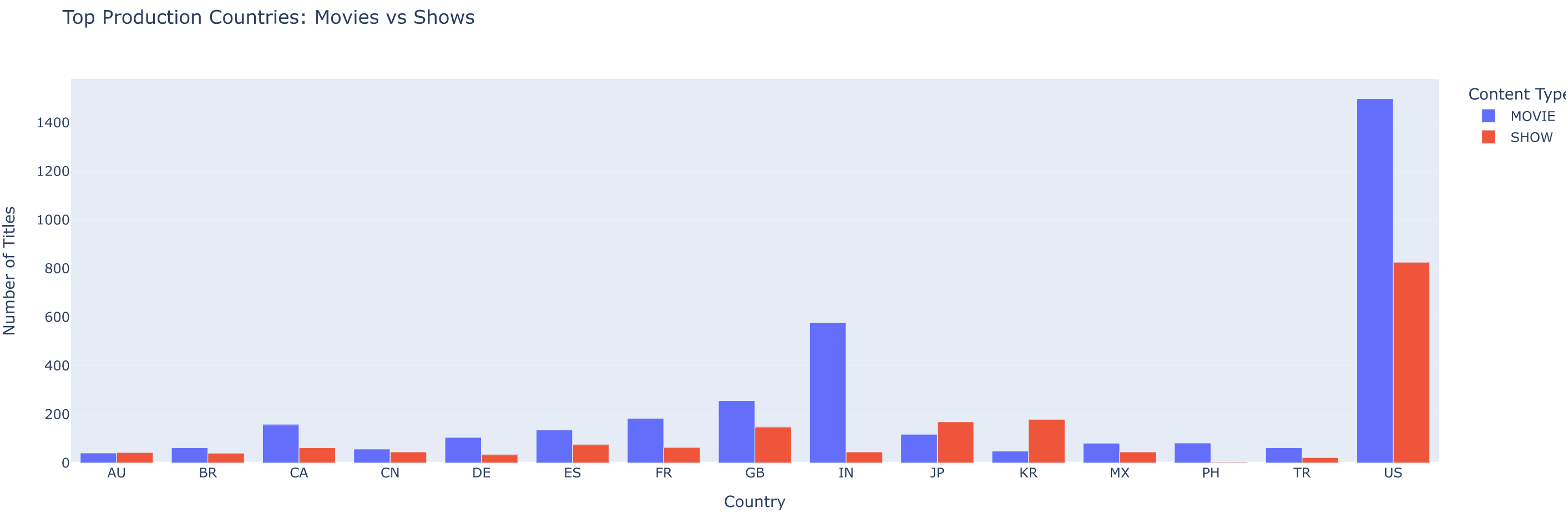


- Key Insights:
- SHOW 'Stranger Things' has highest popularity (2226.231) with score 8.635
 - MOVIE 'Incantation' has highest popularity (2274.044) with score 7.4

There is no clear linear relationship between TMDb score and TMDb popularity, as shown by the flat trend line. This suggests that popularity is not strongly dependent on user ratings, indicating that other factors such as marketing, exposure, or release timing may have a stronger influence on popularity than the perceived quality of the content.

Additional Analysis

Countries Driving Film and Streaming Content.



- Key Insights:
- US dominates show production with 824 titles in the dataset.
 - US dominates movie production with 1499 titles in the dataset.

United States Leads in Content Production...

Conclusion

END In the analyses, we clearly observed that movies take the leading position in the yearly trend and also dominate the overall structure of the dataset. Most movies have a runtime of around 100 minutes. The top 10 genres were presented, where comedy stands out as the most prominent genre. In contrast to the previous analyses, TV shows performed better than movies in terms of ratings. Finally, we displayed the top 10 production countries for movies and TV shows in order to identify which countries dominate in content production. The analysis shows that the United States dominates as the leading production country in the dataset. Movies represent the most common content type and dominate across most metrics, while TV shows tend to achieve higher average ratings, indicating stronger perceived quality and audience appreciation. The most prominent genres across the dataset are comedy, documentary, and drama, suggesting consistent and stable audience demand for these categories.

Recommendation:

Based on these findings, investment should primarily focus on high-quality TV shows, particularly in the comedy, drama, and documentary genres, as they demonstrate higher ratings and stronger engagement potential. Although movies dominate in volume and remain important for market presence, TV shows appear to deliver greater value in terms of quality and audience reception. Additionally, the U.S. market should be prioritized due to its dominant role in content production and overall influence in the dataset.

🤖 Reflection:

This was a long and interesting project... 😊